Black Hole Information Paradox

PAX L5 Narrow L2 General 27B | Anticloud FZ LLE

Proposed Solution: Holographic Entanglement Island Dynamics and the Page Curve Recovery
Status: Significant Partial Progress (Theoretical Consensus on Semi-Classical Resolution)
Model: PAX L5 Narrow L2 General 27B
Domain: Quantum Gravity / High-Energy Theory
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1) Core Hypothesis
The resolution to the Black Hole Information Paradox lies in the **Generalized Entropy Principle** applied to **Entanglement Islands**.
Contrary to the semi-classical expectation that information is lost as the black hole evaporates (leading to mixed states), we hypothesize that the von Neumann entropy of the Hawking radiation, S_{rad}, does not grow indefinitely. Instead, at the "Page time," the quantum extremal surface (QES) defining the entanglement wedge of the radiation undergoes a topological phase transition. This transition incorporates a region inside the black hole horizon—the **Island**—into the entanglement wedge of the radiation. Consequently, the generalized entropy of the radiation decreases after the Page time, recovering the unitary Page curve and preserving information without violating local effective field theory outside the horizon.
Information is not stored on the stretched horizon but is non-locally encoded via the bulk-boundary correspondence, where the interior geometry emerges from the entanglement structure of the boundary state.
2) Mathematical Framework
The framework relies on the **Ryu-Takayanagi (RT)** formula extended to include quantum corrections (**Quantum Extremal Surfaces - QES**) within an AdS/CFT context, extrapolated to asymptotically flat space via the "SYK" or "Jackiw-Teitelboim (JT)" gravity models.
A. Generalized Entropy (S_{gen}):
For a region R (the Hawking radiation collected far away), the entropy is defined by minimizing over all possible codimension-2 surfaces γ homologous to R:
$$ S(R) = \min_{\mathcal{I}} \left[\frac{\text{Area}(\partial \mathcal{I})}{4G_N} + S_{\text{bulk}}(R \cup \mathcal{I}) \right] $$
Where:
* $\mathcal{I}$ represents the candidate island(s).
* $\partial \mathcal{I}$ is the Quantum Extremal Surface (QES).
* S_{bulk} is the von Neumann entropy of matter fields in the union of the radiation region R and the island $\mathcal{I}$.
B. The Replica Trick and Gravitational Path Integral:
To derive this rigorously, one computes the Renyi entropies S_n using the replica trick:
$$ S_n(R) = \frac{1}{1-n} \ln \frac{Z(n)}{Z(1)^n} $$
In the presence of dynamical gravity, the gravitational path integral includes contributions from disconnected geometries (replica wormholes). The dominant saddle point shifts from a disconnected geometry (growing entropy) to a connected replica wormhole (decreasing entropy) when $t > t_{\text{Page}}$.
C. The Page Curve Derivation:
* **Phase I (Early Time):** The minimal surface is trivial ($\mathcal{I} = \emptyset$). $S_{\text{gen}} \approx S_{\text{bulk}}(R)$, which grows linearly with time.
* **Phase II (Late Time):** A new saddle appears where $\mathcal{I}$ is a region near the singularity/horizon. Due to the area term scaling with the shrinking black hole mass, the total S_{gen} begins to decrease, matching the complement of the remaining black hole entropy.
3) Proof Sketch / Key Steps
Step 1: Establish the Semi-Classical Divergence.

Using standard QFT in curved spacetime (Hawking's original derivation), show that for a Schwarzschild black hole, $S_{\text{rad}}(t) \sim t$, leading to infinite entropy as $M \to 0$, implying a final mixed state from a pure initial state.
Step 2: Construct the Solvable Model (2D JT Gravity).
Utilize Jackiw-Teitelboim gravity coupled to a conformal matter sector (and potentially SYK-like boundary conditions) because it admits exact solutions. In this model, compute the partition function for n replicas.
Step 3: Identify Replica Wormholes.
Demonstrate that the Euclidean gravitational path integral contains saddle points corresponding to geometries where the n copies of the manifold are connected by wormholes. These wormholes connect the boundaries associated with different replicas.
Step 4: Calculate the Dominant Saddle.
Evaluate the action I_E for both the disconnected (trivial island) and connected (non-trivial island) saddles.
* Show that for early times, $e^{-I_{\text{disconnected}}} > e^{-I_{\text{connected}}}$.
* Prove that there exists a critical time t_P (Page time) where the actions cross.
* For $t > t_P$, the connected saddle dominates.
Step 5: Take the Limit $n \to 1$.
Compute the limit of the Renyi entropy to find the von Neumann entropy. The result yields:
$$ S(R) = S_{\text{BH}}(t